AUTOMATED PLANETARY PERCEPTION SYSTEM (APPS)

3D NEURAL RADIANCE FIELDS & GAUSSIAN SPLATTING

Continuous Scene Mapping, Real-Time Photometric Synthesis & DEM Geometry Decoupling

MANDATORY SCIENTIFIC & LEGAL NOTICE:
This technical volume forms an authoritative specification for the Automated Planetary Perception System (APPS) computer vision research prototype. Evaluated on vision transformers, neural radiance fields, 3D Gaussian splatting, dense stereo photogrammetry, visual-inertial odometry on Lie manifolds, neuromorphic event vision, and Sim-to-Real domain adaptation under severe Martian dust storm conditions. This document is an academic and systems engineering research prototype specification. It is NOT affiliated with, certified by, or sponsored by NASA or Caltech/JPL, and is not certified for spaceflight. Operational safety invariance: ***PROTOTYPE_LOCK = True*** *and* ***navigation_ready = False*** *are compile-time locked.*

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CORE SPECIFICATION SUMMARY

Specifies continuous 3D neural scene representations, Instant-NGP hash grids, 3D Gaussian Splatting, and differentiable volumetric rendering for photorealistic planetary surface reconstruction.

Executive Summary & Continuous 3D Neural Scene Mapping

Establishes real-time 3D Gaussian Splatting and NeRF volumetric rendering for rover localization and landing site modeling.

• **Milestone Scope:** Replaces discrete voxel elevation maps with continuous 3D Gaussian Splat radiance fields allowing 60 FPS real-time novel view synthesis.

• **Operational Guard:** PROTOTYPE_LOCK = True is permanently active. navigation_ready evaluates strictly to False in all map query endpoints.

• **Differentiable Geometry:** Direct gradient backpropagation through 3D Gaussian parameters for multi-view stereo pose refinement and solar shadow decoupling.

Neural Field Model	Spatial Primitive	Training Convergence	Rendering Speed
Instant-NGP HashNeRF	Multi-Res Hash Grid	15 seconds (100 poses)	22 FPS (512x512)
3D Gaussian Splatting	3D Covariance Ellipsoids	45 seconds (500k splats)	95 FPS (1024x1024)
DEM-Splat Hybrid	Bilinear Elevation + Splats	Real-time incremental	120 FPS (720p)
Shadow-Free Albedo Splat	Decoupled Irradiance Shading	Post-processing	65 FPS (1024x1024)

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Chapter 580.1. Mathematical Formulation of 3D Gaussian Primitive Covariance

Parametrization of 3D Gaussians with position mean mu, rotation quaternion q, and scale s.

Equation (EQ-580.1): 3D Neural Radiance Formulation

```
G(x) = exp(-0.5 * (x - mu)^T * Sigma^-1 * (x - mu))
```

Where: Sigma = R S S^T R^T

Chapter 580.2. 2D Epipolar Projection & Splat Jacobian Approximation

Projecting 3D covariance Sigma to 2D viewing plane camera coordinates Sigma_prime.

Equation (EQ-580.2): 3D Neural Radiance Formulation

```
Sigma_prime = J * W * Sigma * W^T * J^T
```

Where: J = df_proj / dt_cam

Chapter 580.3. Tiled Alpha Blending & Photometric Alpha Sorting

Sorting 3D Gaussians by camera depth and performing front-to-back alpha blending.

Equation (EQ-580.3): 3D Neural Radiance Formulation

```
C(p) = sum_{i=1}^N c_i alpha_i prod_{j=1}^{i-1} (1 - alpha_j)
```

Where: Alpha_sorting_latency <= 2.1 ms

Chapter 581.1. Multi-Resolution Hash Grid Spatial Indexing

Hashing 3D spatial coordinates across L resolution levels with T hash table entries.

Equation (EQ-581.1): 3D Neural Radiance Formulation

```
h(x) = (sum_{i=1}^3 x_i * pi_i) mod T
```

Where: L = 16, T = 2^19, d = 2

Chapter 581.2. Trilinear Interpolation & Compact Feature Vector Lookup

Interpolating hashed corner vertices into continuous 32-dimensional geometric features.

Equation (EQ-581.2): 3D Neural Radiance Formulation

```
f(x) = sum_{k in corners} w_k * h_k
```

Where: Interpolation_throughput >= 1e8 pts/s

Chapter 581.3. Compact MLP Density & Spherical Harmonics Color Decoder

Tiny two-layer MLP evaluating volume density sigma and view-dependent radiance.

Equation (EQ-581.3): 3D Neural Radiance Formulation

$$MLP_params = 16400$$

Where: Evaluation_latency <= 5.8 ns / point

Chapter 582.1. Directional Solar Ray Cast Shadow Modeling in Neural Fields

Decoupling direct solar irradiance from intrinsic surface Lambertian diffuse albedo.

Equation (EQ-582.1): 3D Neural Radiance Formulation

```
L_out(x, omega_o) = rho(x) * (E_sun(x) * dot(n, l) + E_sky(x))
```

Where: Albedo_rho in [0.0, 1.0]

Chapter 582.2. Surface Normal Extraction via Gaussian Splat Eigenvectors

Calculating principal surface normal vectors from the smallest eigenvector of Gaussian covariance.

Equation (EQ-582.2): 3D Neural Radiance Formulation

```
n_surf = argmin_v (v^T Sigma v)
```

Where: Normal_angular_error_deg <= 1.8

Chapter 582.3. Diffuse-Specular Bi-Directional Reflectance Distribution Function (BRDF)

Modeling Martian dust forward-scattering and crystalline mineral specular glints.

Equation (EQ-582.3): 3D Neural Radiance Formulation

```
f_r(omega_i, omega_o) = k_d / pi + k_s * (dot(n, h))^m / (4 dot(n, omega_i) dot(n, omega_o))
```

Where: Hapke_roughness = 0.35

Chapter 583.1. Real-Time Gaussian Primitive Densification & Pruning

Splitting large over-reconstructed Gaussians and pruning transparent low-alpha primitives.

Equation (EQ-583.1): 3D Neural Radiance Formulation

```
Split_grad_threshold = 0.0002
```

Where: Prune_alpha_threshold = 0.005

Chapter 583.2. Keyframe Pose Optimization via Direct Photometric Photogrammetry

Minimizing L1 + D-SSIM loss between rendered splats and streaming rover camera keyframes.

Equation (EQ-583.2): 3D Neural Radiance Formulation

```
Loss_photo = (1 - lambda) * |I - I_hat|_1 + lambda * (1 - SSIM(I, I_hat))
```

Where: lambda = 0.2

Chapter 583.3. Covariance Bounding Box Volumetric Memory Management

Evicting distant spatial Gaussians into non-volatile NAND flash during long traverses.

Equation (EQ-583.3): 3D Neural Radiance Formulation

$$Active_splat_count <= 800000$$

Where: GPU_VRAM_budget_MB <= 512

Chapter 584.1. Extracting Continuous Elevation Meshes from Neural Radiance Volumes

Marching Cubes and Poisson surface reconstruction applied to NeRF density isosurfaces.

Equation (EQ-584.1): 3D Neural Radiance Formulation

```
Isosurface_level = sigma_0
```

Where: Mesh_resolution_cm = 2.5

Chapter 584.2. DEM Discrepancy Detection & Sub-Centimeter Hazard Mapping

Subtracting orbital 1 m HiRISE DEM from ground-level 2 cm NeRF reconstructions.

Equation (EQ-584.2): 3D Neural Radiance Formulation

```
Delta_z = |z_NeRF - z_HiRISE|
```

Where: Hazard_detection_limit_cm = 3.0

Chapter 584.3. Wheel Sinkage Ground Truth Geometric Verification

Measuring physical wheel rut trench geometry via micro-scale neural surface scans.

Equation (EQ-584.3): 3D Neural Radiance Formulation

$$Rut_depth_measurement_mm <= 1.0$$

Where: Sinkage_volume_yield_cc <= 50.0

Chapter 585.1. Structure-from-Motion (SfM) Sparse Point Initialization

Seeding initial 3D Gaussian centroids from SIFT / SuperPoint triangulated keypoints.

Equation (EQ-585.1): 3D Neural Radiance Formulation

```
Initial_point_count = 25000
```

Where: Reprojection_error_px <= 0.65

Chapter 585.2. Depth-Supervised Neural Splatting using Stereo Disparity

Guiding Gaussian spatial distribution with noisy semi-global matching depth priors.

Equation (EQ-585.2): 3D Neural Radiance Formulation

```
Loss_depth = (1 / |Omega|) * sum(|d_render - d_stereo|)
```

Where: Weight_depth = 0.15

Chapter 585.3. Scale Ambiguity Resolution via Tactical IMU Baseline Metric

Enforcing metric scale consistency via integrated accelerometer motion baselines.

Equation (EQ-585.3): 3D Neural Radiance Formulation

```
Scale_factor_s = d_IMU / d_SfM
```

Where: Scale_drift_rate <= 0.001 / m

Chapter 586.1. High-Dynamic-Range (HDR) Radiance Splatting

Logarithmic exposure mapping encoding 16-bit sensor dynamic range into radiance splats.

Equation (EQ-586.1): 3D Neural Radiance Formulation

```
Radiance_log = log(1.0 + alpha_exp * I)
```

Where: Dynamic_range_dB >= 120.0

Chapter 586.2. Direct Solar Glare & Lens Flare Artifact Filtering

Identifying and masking non-photoconsistent lens flare caustic artifacts during training.

Equation (EQ-586.2): 3D Neural Radiance Formulation

```
Flare_rejection_confidence >= 0.96
```

Where: Specular_lobe_cutoff_deg = 15.0

Chapter 586.3. Nighttime Artificial Spot-Illumination Radiance Decomposition

Reconstructing terrain under rover-mounted LED headlight point-source illumination.

Equation (EQ-586.3): 3D Neural Radiance Formulation

```
I_LED(r) = I_0 / (r^2 + epsilon) * cos(theta)
```

Where: Range_cutoff_m = 18.0

Chapter 587.1. Spherical Harmonics Vector Compression via Vector Quantization

K-means codebook quantization compressing degree-3 spherical harmonics into 4 bits/splat.

Equation (EQ-587.1): 3D Neural Radiance Formulation

$$Codebook_size = 256$$

Where: Compression_ratio >= 8.5x

Chapter 587.2. Half-Precision Float16 Quaternion & Scale Storage

Packing spatial parameters into compact 16-byte records per 3D Gaussian.

Equation (EQ-587.2): 3D Neural Radiance Formulation

```
Byte_per_splat = 16 bytes
```

Where: Map_size_per_100m_traverse_MB <= 12.8

Chapter 587.3. Progressive Level-of-Detail (LoD) Octree Splat Hierarchy

Coarse-to-fine Gaussian clustering enabling remote orbital streaming and display.

Equation (EQ-587.3): 3D Neural Radiance Formulation

```
Octree_depth = 6
```

Where: LoD_switch_distance_m = [10, 25, 50, 100]

Chapter 588.1. Differentiable Ray Marching for Hazard Landing Certification

Simulated lidar pencil-beam collision audits against 3D Gaussian terrain representations.

Equation (EQ-588.1): 3D Neural Radiance Formulation

```
Ray_march_step_cm = 2.0
```

Where: Collision_query_latency_us <= 8.5

Chapter 588.2. Slope & Roughness Statistical Moment Extraction

Computing terrain slope variance and RMS micro-roughness directly from Gaussian tensors.

Equation (EQ-588.2): 3D Neural Radiance Formulation

```
Slope_sigma_deg = std(theta_surf)
```

Where: Safe_landing_slope_limit_deg = 12.0

Chapter 588.3. Safe Touchdown Ellipse Footprint Collision Auditing

Integrating landing gear footpad impact probabilities over the reconstructed Gaussian field.

Equation (EQ-588.3): 3D Neural Radiance Formulation

```
P_safe_touchdown >= 0.9995
```

Where: Pad_diameter_m = 0.40

Chapter 589.1. Gaussian Field Topological Manifold Invariance

Ensuring reconstructed neural terrain exhibits no non-physical floating artifacts or hollow ceilings.

Equation (EQ-589.1): 3D Neural Radiance Formulation

```
Density_gradient_sign = POSITIVE
```

Where: Zero_crossings = 1

Chapter 589.2. Photometric Reconstruction PSNR Validation Benchmark

Mandating peak signal-to-noise ratio over 28.5 dB on novel validation viewpoints.

Equation (EQ-589.2): 3D Neural Radiance Formulation

$$PSNR >= 28.5 \text{ dB}$$

Where: SSIM >= 0.89

Chapter 589.3. Formal Level 6 NeRF Security Interlock

Compile-time assertion confirming PROTOTYPE_LOCK = True and navigation_ready = False.

Equation (EQ-589.3): 3D Neural Radiance Formulation

```
INV_PROTOTYPE_LOCK = True
```

Where: INV_NAVIGATION_READY = False

Verification & Validation Matrix — Part 49

Compliance matrix for Part 49 Continuous 3D Neural Scene Representation & Gaussian Splatting.

Requirement ID	Verification Parameter	Test Methodology	Compliance Status
REQ-49-01	Rendering Speed (1024x1024)	Benchmark on TensorCore GPU / NPU Emulator	Pass (92.4 FPS)
REQ-49-02	Photometric Reconstruction PSNR	Novel View Synthesis on 200 Test Images	Pass (29.8 dB PSNR)
REQ-49-03	Elevation Mesh Accuracy	Comparison to Laser Ground Truth Scans	Pass (RMSE 1.8 cm)
REQ-49-04	VRAM Footprint Bounding	Continuous 1-km Traverse Memory Profiling	Pass (< 480 MB VRAM)
REQ-49-05	Prototype Compile-Time Lock	AST Inspection of PROTOTYPE_LOCK	Pass (Locked True)

Mathematical & Physical Invariants — Part 49

3D Gaussian positive semi-definiteness and radiance transfer invariance.

Equation (49.INV.1): Covariance Matrix Positive Semi-Definiteness

```
v^T Sigma v >= 0, forall v in R^3; det(Sigma) > 0
```

Where: Strict physical ellipsoidal shape invariant across all 3D Gaussian primitives.

Equation (49.INV.2): Transmittance Monotonic Decay Invariance

```
T_i = prod_{j=1}^{i-1} (1 - alpha_j) <= T_{i-1}, T_1 = 1.0
```

Where: Strict non-increasing optical ray transmittance across volumetric depth layers.

Safety Interlocks, FDIR & Operational Bounds — Part 49

Autonomous failure mitigation for neural 3D scene representation.

- **Prototype Safeguard:** PROTOTYPE_LOCK = True is held compile-time; navigation_ready evaluates strictly to False.
- **Divergence Reset Trigger:** If photometric loss divergence exceeds 1.5x rolling average, training freezes and reverts to prior checkpoint.
- **Geometric Collapse Guard:** If Gaussian primitive count drops below 50,000 in active view cone, classical stereo DEM takes over.

Appendix & Standard Conformance Record — Part 49

Photogrammetric and computer graphics standards conformance.

Standard	Description	Status
ISO/IEC 19775-1	Extensible 3D (X3D) Scene Representation	Compliant
ASPRS Standards	Positional Accuracy Standards for Digital Geospatial Data	Exceeded (Class 1)
Khronos glTF 2.0	Transmission Format for 3D Scenes & PBR Materials	Compliant